

[별표 1] 인명 구조 및 화재 진압을 위한 항공기 자료

인명 구조 및 화재 진압을 위한 항공기 자료

개 요

이 별표는 인명 구조 및 화재 진압 절차에 대한 원리와 대표적 항공기와 관련하여 화재 진압 및 인명 구조대원에게 필요한 상세한 정보를 제공하기 위한 책자이다.

이 별표의 목적은 항공기에서 효과적인 인명 구조와 화재 진압을 하는데 따른 특정 문제점의 성격을 확실히 파악하는데 필요한 본질적 정보를 인명 구조 및 화재진압 요원에게 제공하는데 있다. 그러나 항공기에 적재된 인화성 액체와 가연성 물질의 양이 항공기의 모델과 운용 면에서 다양하기 때문에, 대표적인 정보만을 제공할 수밖에 없다. 특정 공항에서 항공기를 운용할 때 발생하기 쉬운 갖가지 변화를 파악하기 위해 개별적 검사도 해야 한다.

항공기 제작회사, 항공회사 및 전문기술 서적으로 부터 수집된 자료가 수록되어 있다.

A. PRINCIPLES OF RESCUE

These illustrations show the principal points to consider in gaining access to civil transport aircraft. Each aircraft must be examined individually to know how doors and windows may be most easily opened from outside.

1. LOCATE AND TRY TO GAIN ACCESS AT NORMAL DOORS

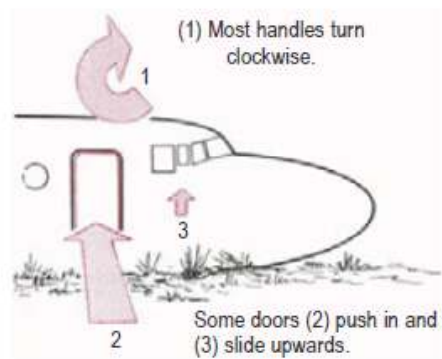
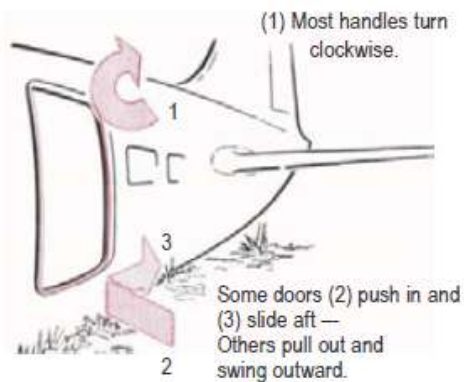
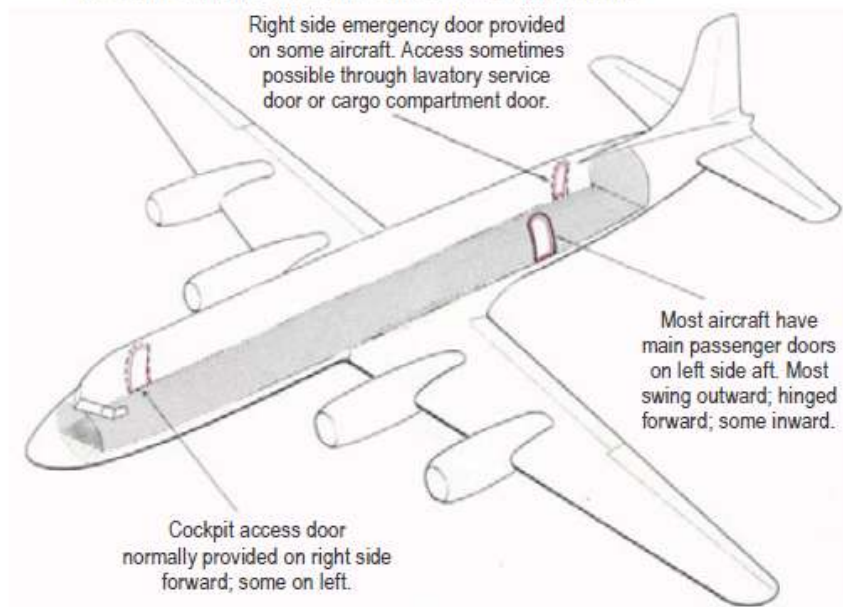


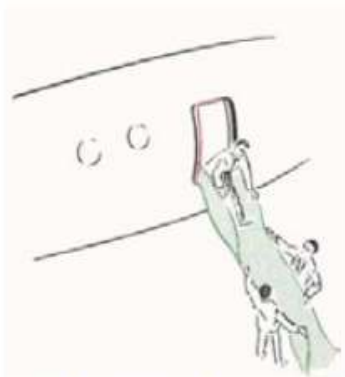
Figure App 1-1. Principles of rescue and firefighting

Some aircraft have integral stairs in nose (as shown) or at the extreme aft end under tail.



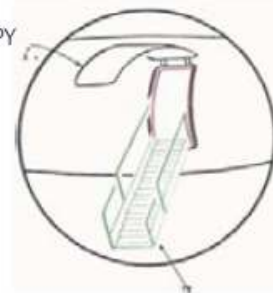
1. PULL "T" HANDLE

2. PULL TWO LATCH HANDLES DOWN AND OUT



Some aircraft have emergency slide escapes at main doors. Some slides must be held at ground level — others are self-inflatable. Passengers should jump into slide.

3. RAISE CANOPY



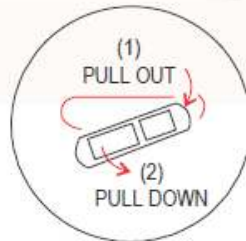
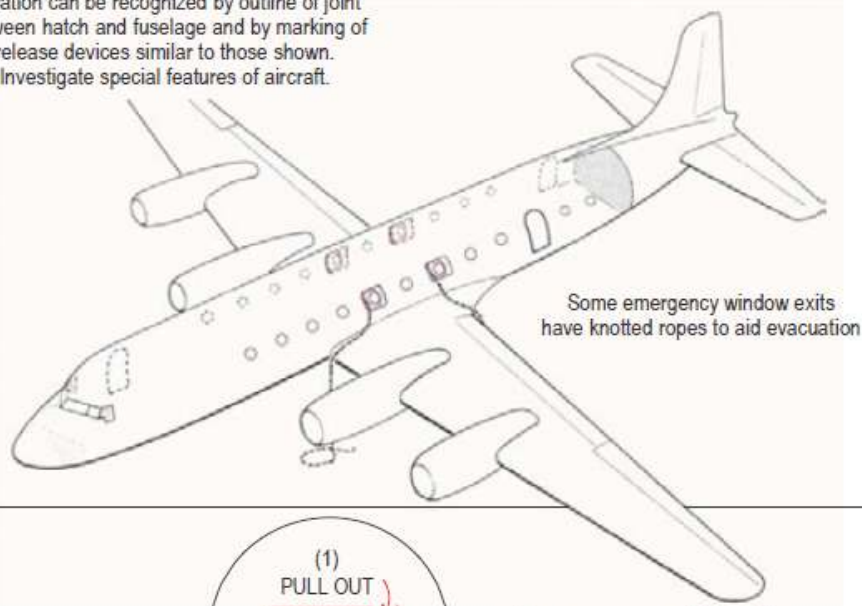
4. PULL DOWN STAIRS

Figure App 1-1. Principles of rescue and firefighting (cont.)

2. LOCATE AND TRY TO OPERATE WINDOW EXITS

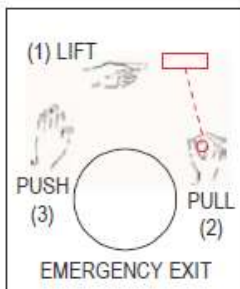
NOTE

Emergency window locations vary.
Location can be recognized by outline of joint
between hatch and fuselage and by marking of
release devices similar to those shown.
Investigate special features of aircraft.



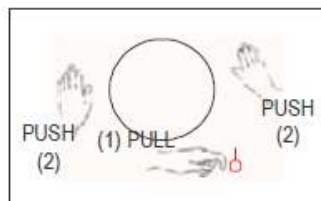
Some emergency window exits have this type red handle — (1) pull out (2) pull down and rotate while still pulling.

Other emergency windows operate like this



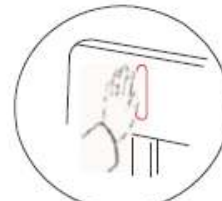
Some windows have bars to LIFT and PULL – then PUSH

Or this



Some windows have rings TO PULL – then PUSH IN

Or this

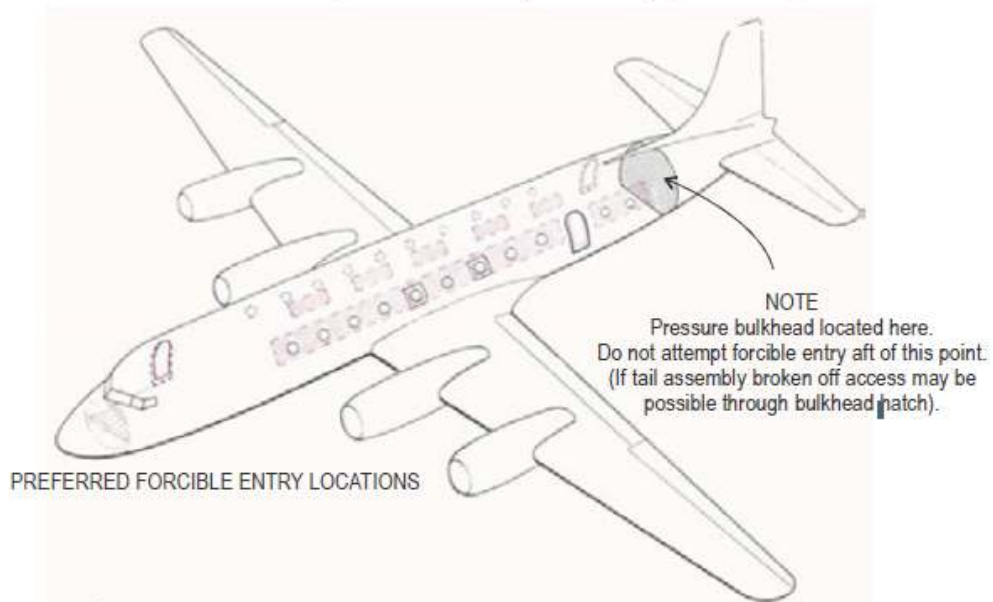


Some windows have BUTTONS TO PUSH

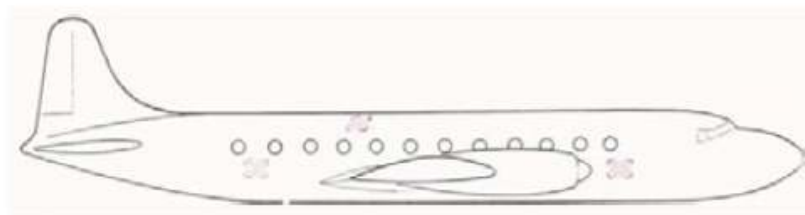
Figure App 1-1. Principles of rescue and firefighting (cont.)

3. AS LAST RESORT MAKE FORCIBLE ENTRY

These illustrations show reciprocating-engine aircraft. Forcible entry locations on modern turbine-powered transport aircraft are most difficult to cut into because of the thickness of the metals used, the extensive framing of the fuselage, the insulation, etc.



1. Force normal or emergency doors or windows if possible.
2. Saw or cut in at or between windows above seat arm level and below the hat rack or on either side of centre line of top fuselage section. Some aircraft have break-in points. Remember when cutting in, occupants may be exposed to injury from cutting tools. Other areas liable to be blocked by internal obstructions.

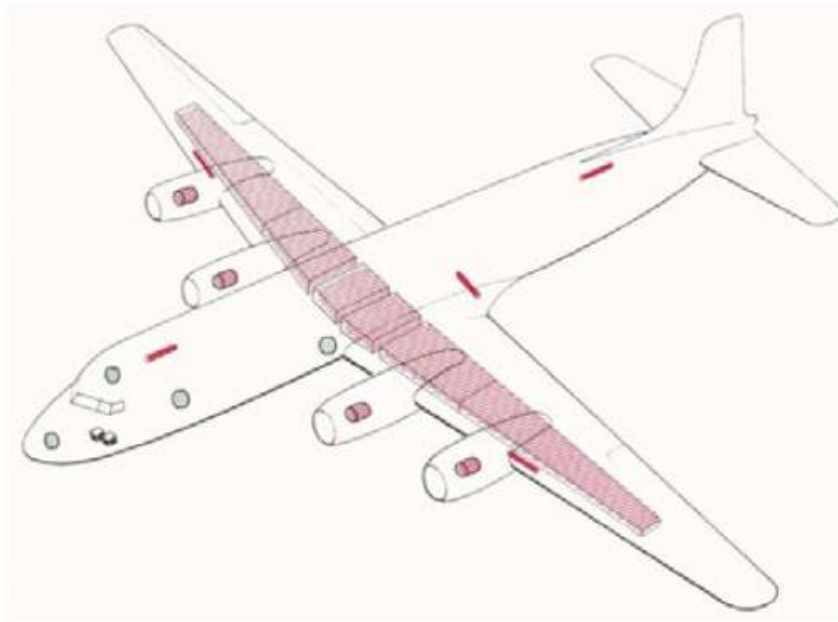


3. Saw or cut in at or break-in points. These points are marked with red or yellow corner markings, and, if necessary, are outlined in white to contrast with the background.

Figure App 1-1. Principles of rescue and firefighting (cont.)

B. PRINCIPAL FIRE HAZARD ZONES IN AIRCRAFT

This is a simplified drawing of the principal fire hazard zones on aircraft.



Fuel tanks normally in wings – some run through fuselage - others all outboard of inboard engines. Fuel tanks are interconnected and have cross-feed valves. Tank vents are normally at trailing edge of wing.



Oil tanks normally in nacelles behind engine firewall – some forward of firewall.



Batteries normally located forward as shown and marked on exterior – disconnect if no fire after crash. Some located in nose wheel well. Quick-disconnect fittings normally are provided.



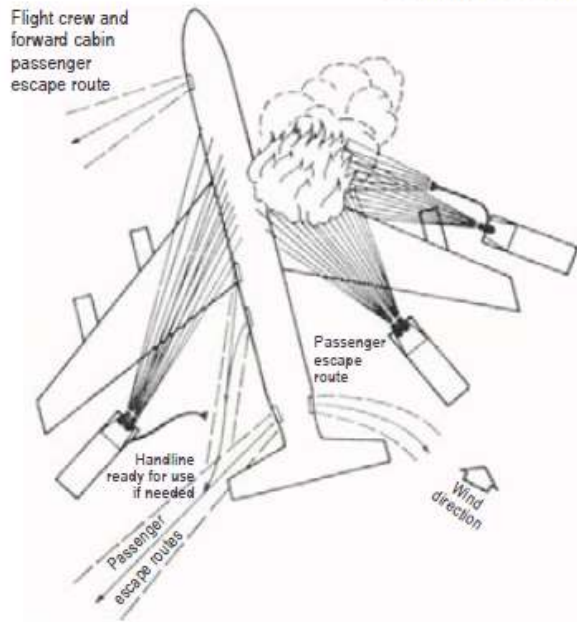
Gasoline combustion heaters located in wings. Fuselage or tail (reciprocating-engine aircraft only).



Hydraulic fluid reservoirs located in fuselage forward or near wing root.

Figure App 1-1. Principles of rescue and firefighting (cont.)

C. PRINCIPLES OF FIREFIGHTING



Using three vehicles dispensing foam with fire in wing-root area on one side and a near crosswind. If port engines are still operating at time firefighting commences, the attack from the port side would have to be shifted forward of the wing.

Using three vehicles with fire involving port inboard engine and integral fuel tank area. Maintaining fuselage integrity is the first principle to allow passenger evacuation.

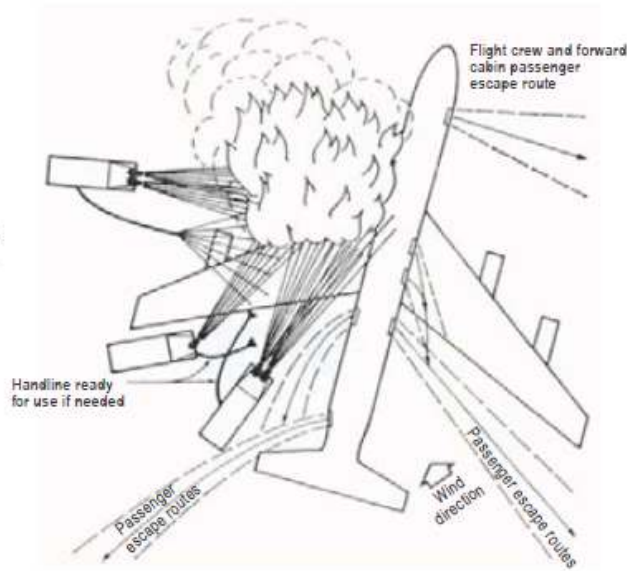
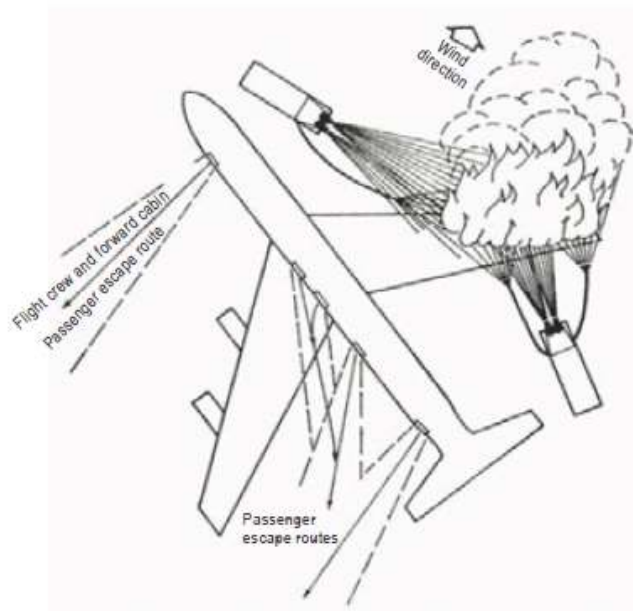


Figure App 1-1. Principles of rescue and firefighting (cont.)



Using two vehicles with fire involving the outboard engine on starboard side. Here the attack is concentrating on controlling the fire and keeping the fuselage shielded from radiant heat and avoiding any direct flame contact which could cause breaching of the fuselage.

Using three vehicles under one of the most adverse conditions where fire involves the entire wingspan of the aircraft. The attack is aimed with the wind and the effort is to keep the fuselage intact while the flight crew and passengers escape via the forward doors.

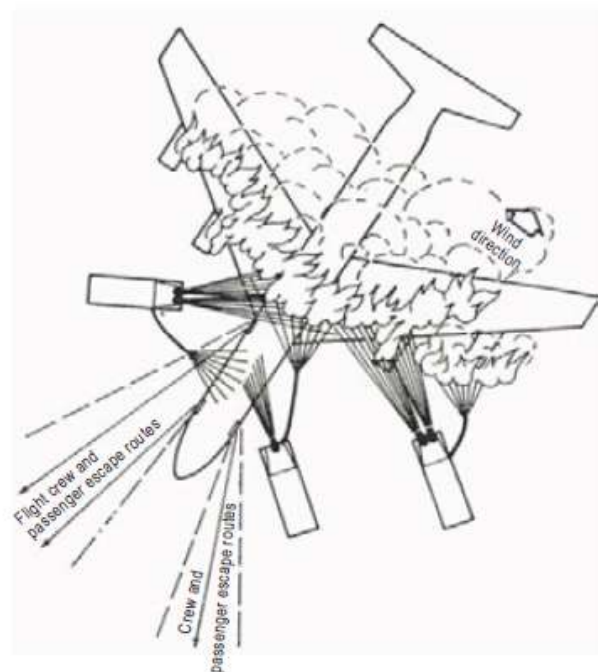


Figure App 1-1. Principles of rescue and firefighting (cont.)